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A.1

2、方程：$\lambda\left(\frac{d^2u}{dt^2}-g\right)-kl\frac{du}{dx}-\frac{mg}{l}=0$

初始条件：$\frac{du}{dt}\Big|_{t=0}$，$u|_{x=0}=k+\frac{H}{l}x$

边界条件：$\frac{du}{dx}\Big|_{x=l}=0$，$u|_{x=0}=0$

3、令$F=\alpha u$，代入扩散方程，

$$\frac{du}{dt}-D\nabla^2 u=\alpha u$$

4、放于球坐标系中，

$$\frac{du}{dn}\Big|_\Sigma=-H(u_0-u|_\Sigma)+M\frac{d\Sigma}{dn}$$

$$\Rightarrow\left(\frac{du}{dr}+Hu\right)\Big|_\Sigma=M\cos\theta \qquad \Sigma: r=a$$